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# Project Documentation

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## 1. Introduction

**Project Title:** “OptiCrop – Smart Agricultural Production Optimization Engine”

**\*Team Members**

| Team Member      | Role        |
|------------------|-------------|
| K.Sri Bala       | Team Lead   |
| P.Umamaheswari   | Team Member |
| P.Siva Durga     | Team Member |
| S.Charukesh Babu | Team Member |
| T.Sumanth        | Team Member |

## 2. Project Overview

**Purpose:**

OptiCrop is a machine learning-based agricultural recommendation system designed to optimize crop production using environmental and soil parameters. It integrates ML algorithms with a Flask web application to provide intelligent crop

recommendations based on key environmental factors, helping farmers, agricultural researchers, agribusiness companies, and policymakers make data-driven decisions that improve productivity, sustainability, and resource utilization.

#### Features:

- Smart crop recommendation based on soil nutrients (N, P, K) and environmental parameters.
- Crop suitability and environmental assessment for a given set of conditions.
- Support for agricultural research and policy planning through data-driven pattern analysis.
- Comparison of multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means) to select the best-performing model.
- Interactive, responsive web interface for entering parameters and viewing instant predictions.
- Trained model persistence using Pickle (.pkl) for fast reuse during prediction.